CLASS 9 · JEE MAINS · INSTITUTE TOP 10 VS OUR TOP 10

Pragya G9 (EXT) Achievers

Part Test-1 · where our toppers fall behind the institute's toppers

Exam date 08 Oct 2026
Campus HYD/Nallakunta DD Colony
Compared institute top 10 vs our top 10

On average the institute's top 10 scored **203.3** and our top 10 scored **195.5** – a gap of **7.8 marks** per student: **2.1** in Maths, **5.0** in Physics and **0.7** in Chemistry.

Maths 27% of the gap | Physics 64% of the gap

7 of the institute's top 10 are our students (program rank 1, 2, 3, 4, 6, 9, 10), so this compares the institute's other 3 toppers with our next 3. Scores are rebuilt from each question's right / wrong / skipped counts with the paper's marking (our average matches the official 195.5 exactly).

203 / 296
institute top 10 – average score

196 / 296
our top 10 – average score

8 questions
where 2+ more of the institute top 10 got it right

3 questions
where 2+ more of ours got it right

1 Subject by subject

Average marks per student, and how the answers split into right, wrong and skipped. ■ Institute top 10 ■ Our top 10

SUBJECT	AVERAGE SCORE		GAP	RIGHT	WRONG	SKIPPED
Maths out of 100		76.1 74.0	2.1 per student	80% vs 78%	16% vs 18%	4% vs 4%
Physics out of 100		61.4 56.4	5.0 per student	67% vs 62%	22% vs 24%	12% vs 14%
Chemistry out of 96 = 100 minus Q58 (4)		65.8 65.1	0.7 per student	73% vs 73%	18% vs 20%	10% vs 7%

"Out of" = the subject's marks on this paper minus the questions left out by the final key (marks of each left-out question in brackets). Percentages are of all answers in that subject (questions × students), institute first, ours second.

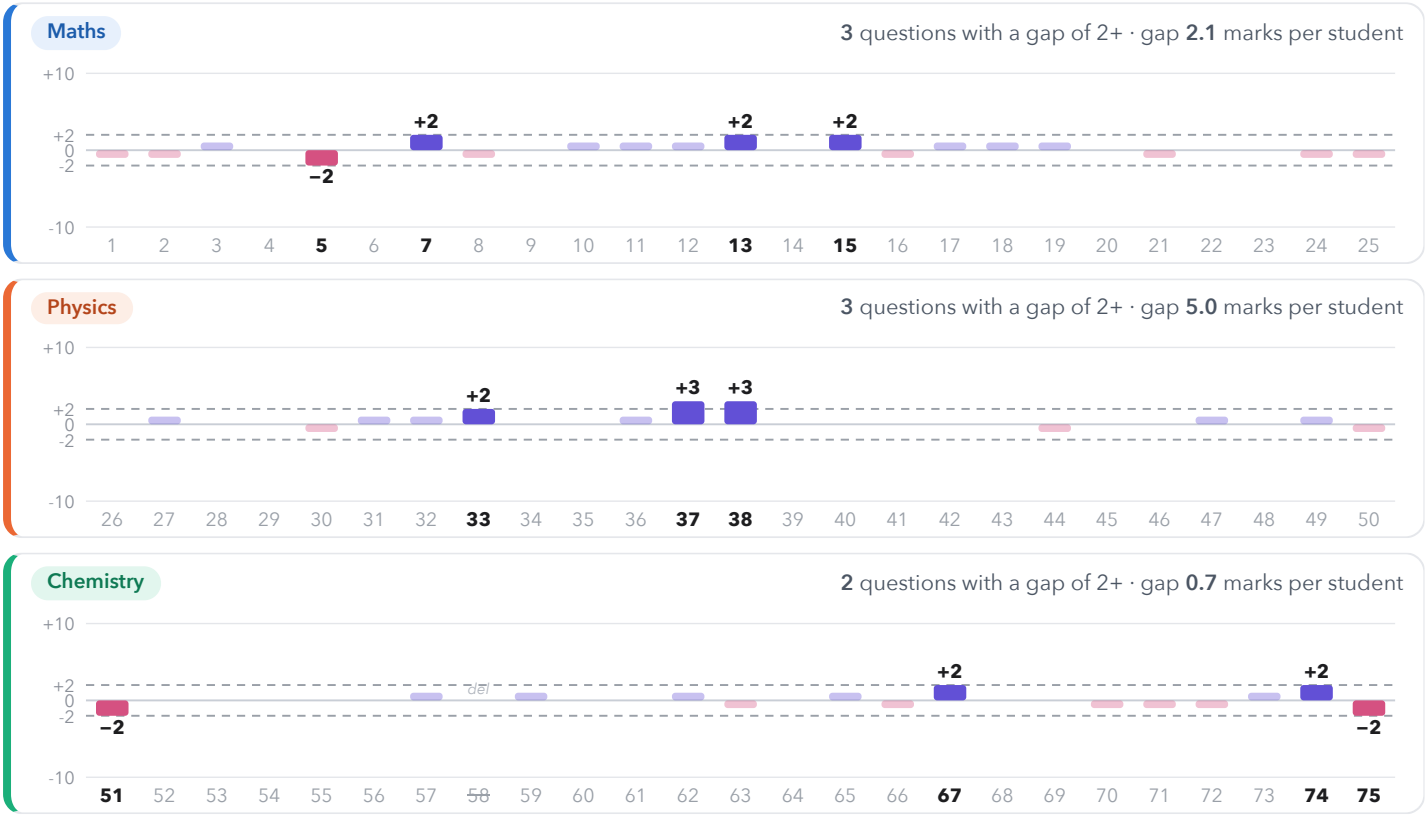
Key takeaways

1. **7 of the institute's top 10** are our own students (program rank 1, 2, 3, 4, 6, 9, 10), so the two groups mostly overlap: at most 3 answers can differ on any question, so the gap line is set at 2 instead of 4.
2. **Physics** is where we fall furthest behind: **5.0** marks per student (64% of the total gap).
3. **8 questions** were answered correctly by at least 2 more of the institute's top 10 than ours. Biggest: Q37 (8 vs 5), Q38 (7 vs 4), Q7 (5 vs 3).
4. Where we did better: Q5 (9 vs 7), Q51 (9 vs 7), Q75 (4 vs 2).

2 Question-by-question gap

For each question: how many more of the institute's top 10 got it right than ours. Bars above the line = the institute did better; below = we did better. Gaps of 2+ (dashed line) are labelled.

Institute top 10 got more right · Our top 10 got more right · ---- Gap of 2



3 Chapter gaps

Average marks per student on each chapter's questions, institute top 10 vs ours. Subjects are ordered from the biggest gap to the smallest; within each subject, largest chapter gap first.

CHAPTER	QS	INSTITUTE VS OURS	GAP PER STUDENT	GAP QUESTIONS
1. Physics gap 5.0 marks per student · 3 questions with a gap of 2+				
Laws of Motion	8	23.7 vs 20.2	3.5	Q33, Q37
Work, Energy & Power	5	10.0 vs 7.9	2.1	Q38
Circular Motion	7	17.1 vs 17.1	0.0	–
Friction	2	3.7 vs 3.7	0.0	–
Momentum & Collisions	3	6.9 vs 7.5	−0.6	–
2. Maths gap 2.1 marks per student · 3 questions with a gap of 2+				
Trigonometric Equations	7	22.7 vs 21.3	1.4	Q7, Q13
Inverse Trigonometric Functions	10	31.5 vs 30.7	0.8	–
Sequences & Series	1	4.0 vs 3.6	0.4	–
Functions	7	17.9 vs 18.4	−0.5	Q15
3. Chemistry gap 0.7 marks per student · 2 questions with a gap of 2+				
General Organic Chemistry	10	26.2 vs 24.1	2.1	Q67, Q74
Isomerism	10	31.8 vs 32.3	−0.5	–
Hydrocarbons	4	7.8 vs 8.7	−0.9	–

4 Biggest gaps – question list

Questions where at least 2 more of the institute's top 10 got the right answer than ours, one table per subject. The cropped question for each is in section 5. With 7 of the 10 students shared, at most 3 answers can differ on any question, so the line is set at 2 instead of 4.

Maths

3 questions with a gap of 2+

Q	CHAPTER – WHAT IT TESTS	TYPE	INSTITUTE RIGHT	OURS RIGHT · WRONG · SKIPPED	GAP
Q7	**Trigonometric Equations** – $\sqrt{3}\sin \pi x + \cos \pi x =$ quadratic – number of solutions	Single correct · +4 / −1	5/10	3/10 · 6 · 1	+2
Q13	**Trigonometric Equations** – Solutions of $\tan x + \sec x = 2\cos x$ in $[0, 2\pi]$	Single correct · +4 / −1	10/10	8/10 · 1 · 1	+2
Q15	**Functions** – Inverse of $f(x) = x^2 + x + 1$ on a restricted domain	Single correct · +4 / −1	10/10	8/10 · 1 · 1	+2

Physics

3 questions with a gap of 2+

Q	CHAPTER – WHAT IT TESTS	TYPE	INSTITUTE RIGHT	OURS RIGHT · WRONG · SKIPPED	GAP
Q37	**Laws of Motion** – Tension in a pulley–block system	Single correct · +4 / −1	8/10	5/10 · 2 · 3	+3
Q38	**Work, Energy & Power** – Work along different paths; conservative or not	Single correct · +4 / −1	7/10	4/10 · 3 · 3	+3
Q33	**Laws of Motion** – Tangential acceleration on a frictionless curved track	Single correct · +4 / −1	8/10	6/10 · 1 · 3	+2

Chemistry

2 questions with a gap of 2+

Q	CHAPTER – WHAT IT TESTS	TYPE	INSTITUTE RIGHT	OURS RIGHT · WRONG · SKIPPED	GAP
Q67	**General Organic Chemistry** – Structure and hybridisation of the methyl carbocation	Single correct · +4 / −1	8/10	6/10 · 2 · 2	+2
Q74	**General Organic Chemistry** – Compounds giving a positive bicarbonate test	Numerical · +4 / −1	7/10	5/10 · 3 · 2	+2

Where our top 10 did better

Q5 9 vs 7 right · **Q51** 9 vs 7 right · **Q75** 4 vs 2 right

How to read this report. "Institute top 10" = the *Top 10 Students* rows of the institute's question-wise R/W/U sheet for this test (program ranks 1–10 in Pragya G9 (EXT) Achievers, all campuses). "Our top 10" = this campus's top 10 by program rank. A gap of +5 on a question means 5 more of the institute's top 10 got it right than ours.

5 Maths – biggest gaps

3 questions, largest gap first
cropped from the question paper

Q7 — Trigonometric Equations – √3 sin πx + cos πx = quadratic – number of solutions

Gap +2

Institute top 10: **5 right**, 4 wrong, 1 skipped · Our top 10: **3 right**, 6 wrong, 1 skipped
Maths · Single correct · marking +4 / −1

7. Number of solutions of $\sqrt{3}\sin(\pi x)+\cos(\pi x)=x^2-\frac{2}{3}x+\frac{19}{9}$ is .....

A) 0 B) 1 C) 2 D) 3

Ours wrong: Masarapu Kartikeya, Ramini Havisha, Dharavath Joseph, Pulipati Vikas, Rachoori Vishwateja, Vyasabhattu Shreyasraju

Ours skipped: Nandimandalam Loukhya Reddy

Q13 — Trigonometric Equations – Solutions of tan x + sec x = 2 cos x in [0, 2π]

Gap +2

Institute top 10: **10 right**, 0 wrong, 0 skipped · Our top 10: **8 right**, 1 wrong, 1 skipped
Maths · Single correct · marking +4 / −1

13. Number of solution(s) of $\tan x+\sec x=2\cos x$ in $[0,2\pi]$ is

A) 1 B) 2 C) 3 D) 4

Ours wrong: Pulipati Vikas

Ours skipped: Rachoori Vishwateja

Q15 — Functions – Inverse of f(x) = x² + x + 1 on a restricted domain

Gap +2

Institute top 10: **10 right**, 0 wrong, 0 skipped · Our top 10: **8 right**, 1 wrong, 1 skipped
Maths · Single correct · marking +4 / −1

15. Let $f:\left(-\infty,\frac{-1}{2}\right]\to\left[\frac{3}{4},\infty\right)$, $f(x)=x^2+x+1$. Then $f^{-1}(x)=$ .....

A) $\frac{-1\pm\sqrt{4y-3}}{2}$ B) $\frac{-1\pm\sqrt{4x-3}}{2}$

C) $\frac{-1+\sqrt{4x-3}}{2}$ D) $\frac{-1-\sqrt{4x-3}}{2}$

Ours wrong: Pulipati Vikas

Ours skipped: Rachoori Vishwateja

Physics – biggest gaps

3 questions, largest gap first
cropped from the question paper

Q37

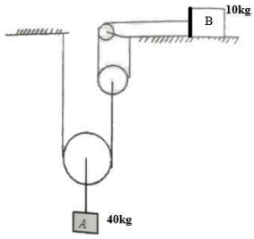
Laws of Motion – Tension in a pulley-block system

Institute top 10: **8 right**, 1 wrong, 1 skipped · Our top 10: **5 right**, 2 wrong, 3 skipped

Physics · Single correct · marking +4 / −1

Gap +3

37. Fig shows two blocks A and B connected to an ideal pulley string system. In this system when bodies are released then $\left(g = 10m/s^2\right)$(Mass of a is 40Kg and B is 10Kg on smooth surface). The tension in the string connecting the block B is



A) $20N$ B) $80N$ C) $60N$ D) $40N$

Ours wrong: Dharavath Joseph, Rachoori Vishwateja

Ours skipped: Ramini Havisha, Pulipati Vikas, Vyasabhattu Shreyasraju

Q38

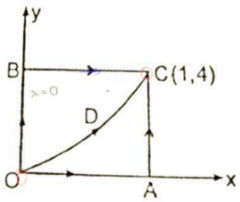
Work, Energy & Power – Work along different paths; conservative or not

Institute top 10: **7 right**, 0 wrong, 3 skipped · Our top 10: **4 right**, 3 wrong, 3 skipped

Physics · Single correct · marking +4 / −1

Gap +3

38. A particle is moved along the different paths OAC, OBC and ODC as shown in the fig. path ODC is parabola, $y = 4x^2$. The work done by a force $\vec{F} = xy\hat{i} + x^2y\hat{j}$ on the particle along



Column – I	**Column – II**
a)Path OAC	P) 2J
b) Path OBC	Q) $\frac{19}{3}$J
c) Path ODC	R) Conservative
d) Force is	S) 8J
	T) Non- conservative

A) $a \to Q, b \to S, c \to P, d \to T$ B) $a \to S, b \to P, c \to Q, d \to T$

C) $a \to S, b \to Q, c \to P, d \to T$ D) $a \to S, b \to Q, c \to S, d \to T$

Ours wrong: Pulipati Vikas, Rachoori Vishwateja, Vyasabhattu Shreyasraju

Ours skipped: Puneeth Karthikeya Jureddi, Nandimandalam Loukhya Reddy, Masarapu Kartikeya

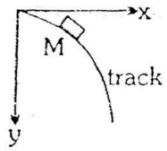
Q33 **Laws of Motion – Tangential acceleration on a frictionless curved track**

Institute top 10: **8 right**, 0 wrong, 2 skipped · Our top 10: **6 right**, 1 wrong, 3 skipped

Physics · Single correct · marking +4 / −1

Gap +2

33. A man M slides down a curved frictionless track, starting from rest. The curve obeys equation $y = \frac{x^2}{2}$. The tangential acceleration of mass is



A) g B) $\frac{gx}{\sqrt{x^2+4}}$ C) $\frac{g}{2}$ D) $\frac{gx}{\sqrt{x^2+1}}$

Ours wrong: Vyasabhattu Shreyasraju

Ours skipped: Nandimandalam Loukhya Reddy, Ramini Havisha, Pulipati Vikas

Chemistry – biggest gaps

2 questions, largest gap first
cropped from the question paper

Q67

General Organic Chemistry – **Structure and hybridisation of the methyl carbocation**

Institute top 10: **8 right**, 0 wrong, 2 skipped · Our top 10: **6 right**, 2 wrong, 2 skipped

Chemistry · Single correct · marking +4 / −1

Gap +2

67. Wrong statement regarding methyl carbonium ion $\overset{\oplus}{C}H_3$

A) It is sp^2 hybridized

B) Vacant orbital is sp^2 hybridized

C) vacant orbital is perpendicular to molecular planar and is pure p-orbital

D) It is electrophilic with sextet of electron

Ours wrong: Pulipati Vikas, Vyasabhattu Shreyasraju

Ours skipped: Nandimandalam Loukhya Reddy, Masarapu Kartikeya

Q74

General Organic Chemistry – **Compounds giving a positive bicarbonate test**

Institute top 10: **7 right**, 0 wrong, 3 skipped · Our top 10: **5 right**, 3 wrong, 2 skipped

Chemistry · Numerical · marking +4 / −1

Gap +2

74. How many of the following compounds can give positive bicarbonate test?

HCl, Picric Acid, Benzoic acid, 2,4-dintrophenol, 4-nitrophenol, Squaric Acid, formic Acid, Benzene Sulphuric acid, 1-Butyne, NH_4Cl .

Ours wrong: Pulipati Vikas, Rachoori Vishwateja, Vyasabhattu Shreyasraju

Ours skipped: Ramini Havisha, Dharavath Joseph